William Steinley

Forester / GIS Analyst · M.S. Forestry

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PROFESSIONAL SUMMARY

Forester and Geographic Information Systems (GIS) analyst with 8+ years in forestry, including 2+ years of GIS and Light Detection and Ranging (LiDAR) analysis, an M.S. in Forestry, and California Registered Professional Forester (RPF) exam eligibility. Timber sale layout and cruising in seven states and federal timber sale work on the Plumas National Forest. ArcGIS Pro, Field Maps, and Survey123 for field data; R and Python for LiDAR. Lead author of a journal article from the M.S. thesis on fire return intervals.

TECHNICAL SKILLS

Forestry and Field Operations: Timber Cruising (Fixed-Plot, Variable-Radius, Strip, 100%), Silvicultural Prescription Implementation, Logging Systems (Cable, Helicopter, Tractor), Forest Inventory, Cruise Compilation, GPS Data Collection, Timber Sale Layout, Harvesting Operations, Timber Stand Improvement and Crop Tree Release, Post-Harvest and Regeneration Surveys, Boundary Marking and Painting, Best Management Practices (BMP) and Streamside Management Zones (SMZ), Tree and Shrub ID (Dendrology)
Fire, Vegetation, and Management: Prescribed Fire and Fuels Reduction, Vegetation Management, Crew Supervision and Quality Control, National Environmental Policy Act (NEPA) Implementation, Fireline Construction, Wildfire Mitigation, Invasive Species Control, Habitat Restoration, Chainsaw Operation, ATV and Heavy Equipment (D5 Dozer), Project Management, Natural Resource Management, Landowner Outreach, Contractor Coordination
GIS and Spatial Science: ArcGIS Pro 3.x, ArcMap, QGIS 3.x (PyQGIS), ArcGIS Online Publishing and Management, Esri Field Maps, Survey123, Spatial Analysis, Remote Sensing (LiDAR, aerial imagery), LiDAR Processing (lidR, PDAL, GDAL; Canopy Height Model [CHM], Digital Terrain Model [DTM]), Geodatabase Design and Management, Cartographic Production, Slope and Terrain Modeling
Programming and Data Management: R (lidR, terra, sf, tidyverse), Python (ArcPy, pandas, GeoPandas), Rust, Statistical Analysis, Microsoft Office (Word, Excel, Outlook, PowerPoint), Git and GitHub, Automated Workflow Development, Technical Writing and Documentation
Field Equipment and Software: Clinometer, Relaskop, Increment Borer, Laser Rangefinder, Logger's Tape, Compass, Densiometer, Handheld Data Recorders, Avenza Maps, onX, Google Earth, Google Workspace

WORK EXPERIENCE

Principal / Consulting Forester

Mar 2026 – Present · Self-employed, hours vary

Athena Resource Group, Gold Run, CA

- Write land management plans for private landowners. Each plan turns owner objectives into a written prescription and a treatment schedule. Part-time consulting practice with field projects in Texas.
- Carry out the prescribed treatments: chainsaw and hand thinning, brush and invasive species control.
- Run the business side of the practice: client acquisition, project scoping, and invoicing.

Forester III

Feb 2024 – Mar 2026 · Full-time, 50 hrs/wk

Jefferson Resource Company (JRC), Placerville, CA

- Supervised field crews of up to four personnel across ~8,000 acres on the Plumas National Forest Community Protection Project. Reviewed 500+ plots and caught layout and survey errors before U.S. Forest Service (USFS) deliverables went out.
- Applied NEPA requirements and silvicultural prescriptions in the field, including layout of tractor, cable and helicopter yarding units.
- Led the move from Avenza Maps Pro to ArcGIS Online, ArcGIS Pro, Esri Field Maps, and Survey123. Crews stopped re-keying field data and got real-time cloud access.
- Wrote two technical guides and the training materials that brought the crews over.
- Automated map production and spatial data processing in Python and ArcPy, replacing hand-assembled maps with one standard output across deliverables. Validated every output against manual quality-control checks.
- Built CHM and DTM products from LiDAR point clouds in R (lidR), checking point density and resampling before setting grid resolution. Derived slope and terrain layers used to pick yarding systems and delineate harvest units.
- Wrote R and lidR routines for individual tree crown delineation and canopy cover estimation, an open-source alternative to licensed processing software.
- Published and maintained 40+ hosted feature layers and geodatabases, with offline map areas, across three project areas. Crews used them for stream, inventory plot, archaeological, and forest disease and dieback data.
- Field liaison between JRC, USFS timber staff, and National Forest Foundation partners. Supported contract administration and USFS best management practice compliance under a Registered Professional Forester.

Forester
Apr 2022 – Jul 2023 · Full-time, 50 hrs/wk

Sunrise Pines Forestry, Flagstaff, AZ (projects in seven states)

- Laid out timber sales in seven states, from Rocky Mountain mixed conifer to Gulf Coastal Plain pine. Ground-based systems throughout, cable systems on steep ground in California and Montana.
- Cruised timber by fixed-plot, variable-radius, strip, and 100% methods across forest types and ownerships, including 250+ plots in eastern Oklahoma. Compiled volume summaries for appraisals, management plans, and harvest scheduling.
- Marked timber to silvicultural prescriptions in stands from ponderosa pine and mixed conifer to bottomland hardwood and shortleaf pine. Worked out implementation with landowners, consulting foresters, and logging contractors.

Fuels Technician
Jan 2022 – Apr 2022 · Full-time, 40 hrs/wk

Fire Safe Council of Nevada County, Grass Valley, CA

- Carried out fuel reduction prescriptions under a Registered Professional Forester with chainsaws, tow-chippers, tracked masticators, and hand tools for community wildfire protection. Moved up to the Sunrise Pines forester role after three months.
- Constructed control lines to burn plan specifications and worked prescribed fire ignition, holding, and mop-up operations.

Graduate Teaching Assistant
Aug 2019 – Jan 2022 · Part-time, 20 hrs/wk

Stephen F. Austin State University, Nacogdoches, TX

- Taught lab sections of 30–50 undergraduate students per semester in Introduction to Forestry and Forest Biometrics. Led field trips and taught mensuration, sampling design, and spatial analysis.
- Assisted faculty at the six-week summer field station program. Hauled and staged field equipment and kept accountability for 100+ students through exercises in timber cruising, silviculture, and forest management planning.

Fuels Technician
Nov 2017 – May 2019 · Part-time, 20 hrs/wk

TimberPro Forest Service, Nacogdoches, TX

- Operated D5 bulldozer to construct control lines and suppress slopovers and spot fires during prescribed burn operations across East Texas pine plantations.
- Ignited prescribed burns with drip torches and inspected control lines by ATV. Suppressed spot fires with hand tools across multiple burn units.

SELECTED PROJECTS

- **Mohawk Valley West Slope harvest planning (2026, independent project on public data):** USGS 3DEP QL1 LiDAR to 3 ft DTM, CHM, and planning-slope rasters (PDAL, GDAL); 24 demonstration units, 1,621 gross ac; skyline screen of 8,540 corridors from 314 candidate landings, PNW-39 payload estimates; 25-sheet 11x17 PyQGIS map series, GeoPDFs for Avenza; simulated BAF 20 cruise checked to FSH 2409.12.
- **Forest Inventory Analyzer (Rust, v0.2.0):** cruise compiler with Survey123/Field Maps-style import, BAF expansion, TPA, basal area, QMD, diameter-class tables, Student's t sampling error and growth projections; CSV/JSON/Excel input; CI-tested releases.

EDUCATION

Master of Science in Forestry
2024

Stephen F. Austin State University

Thesis: *Consequences of Altered Burn Regimes on Plant Community Diversity in the Pineywoods of East Texas*

Honors: Xi Sigma Pi Forestry Honor Society

Thesis research and writing: Jul 2023 – Feb 2024

Bachelor of Science in Forestry
2019

Stephen F. Austin State University

Minor: Geospatial Science · Concentration: Fire Management

Relevant Coursework: Forest Biometrics, GIS and Remote Sensing, Fire Ecology, Spatial Science, Silviculture

CERTIFICATIONS

- **ELIGIBLE**: California Registered Professional Forester (RPF), approved by the Board of Forestry and Fire Protection to sit for the exam
- **ELIGIBLE**: ISA Certified Arborist, approved by the International Society of Arboriculture to sit for the exam
- Qualified Cruiser (FSH 2409.12 ch. 60), U.S. Forest Service, Region 3, certified 2022–2023 while cruising for Sunrise Pines Forestry
- Tree Risk Assessment Qualification (TRAQ), International Society of Arboriculture, current
- CPR Training, current
- Valid California Driver's License, clean record; own 4WD vehicle

PUBLICATIONS

Steinley, W. J., Oswald, B. P., Kidd, K. R., & Glasscock, J. (2024). *Influence of altered fire return intervals on plant diversity in the East Texas Pineywoods.* Forest Research: Open Access, 13(5), 529.